Universitatea Transilvania din Brașov

ROVISlaboratory

FIERY: Future Instance Prediction in Bird's-Eye View from Surround Monocular Cameras

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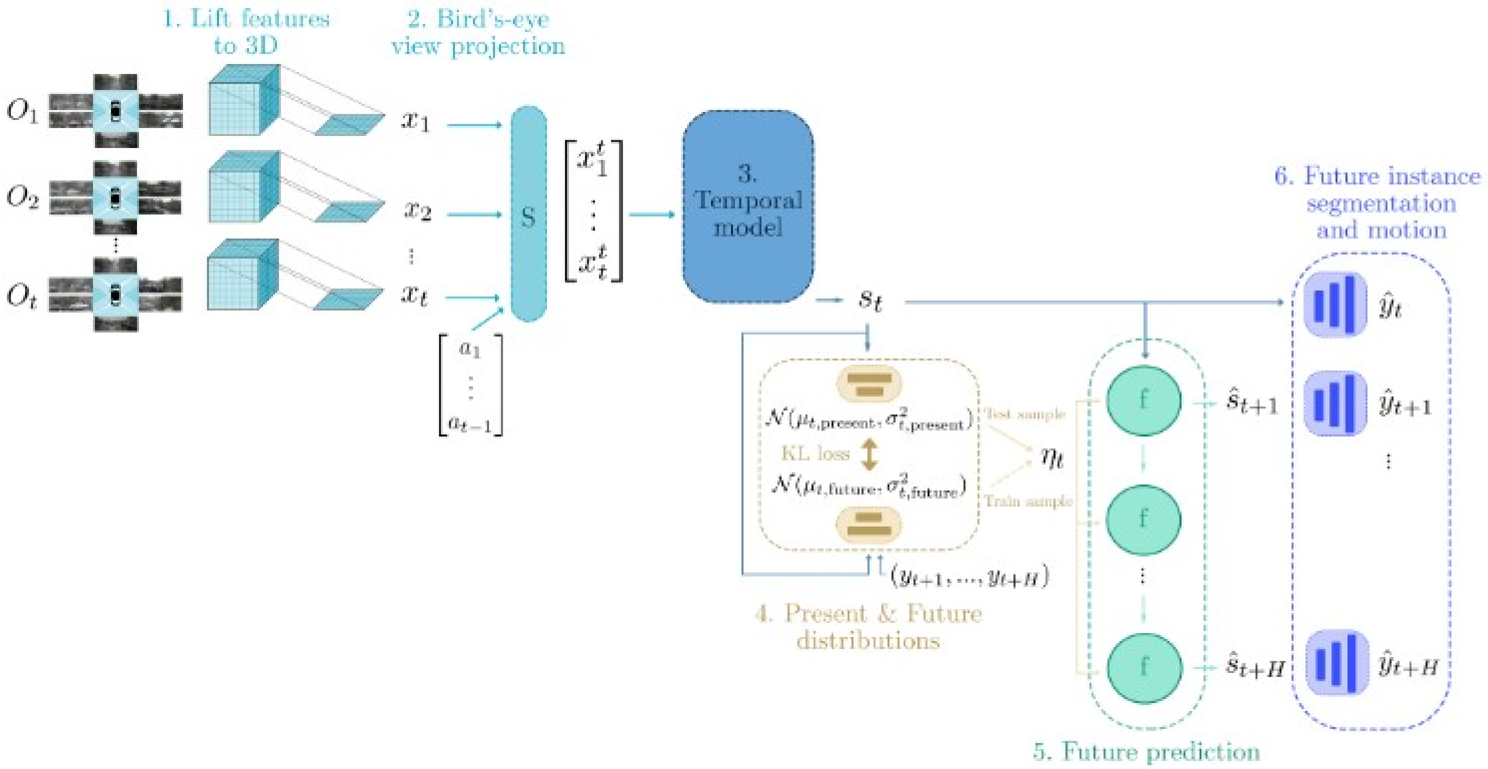
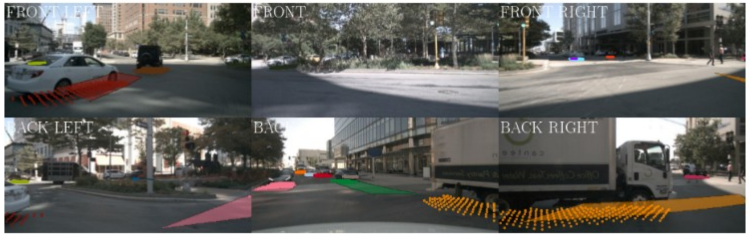
FIERY Architecture



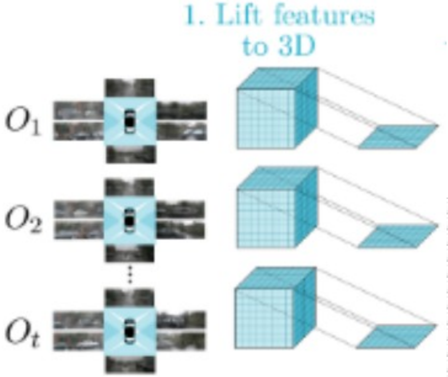
1. Lift Features to 3D

At each past timestep {1, …, t}, we lift camera inputs (O_1, …, O_t) to 3D by predicting a depth probability distribution over pixels and using known camera intrinsics and extrinsics

Backbone	EfficientNet-B4, stride 8
Feature spatial size	$(H_e, W_e) = (28, 60)$
Feature channels C	64
Depth bins D	48 (Dmin=2m, Dmax=50m, Dsize=1m)
BEV grid	200×200 at 0.5m resolution
Capture area	100m × 100m around ego-vehicle
BEV feature	$x_t \in \mathbb{R}^{64\times200\times200}$



(a) Camera inputs.



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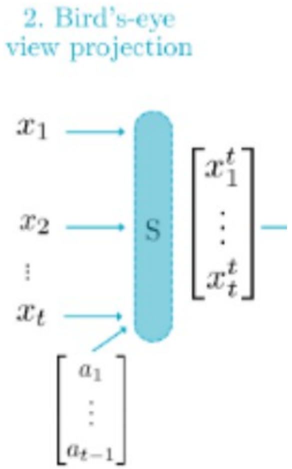
2. Bird’s-Eye View Projection

These features are projected to bird’s-eye view (x_1, ..., x_t). Using past ego-motion (a_1, ..., a_{t-1}), we transform the bird’s-eye view features into the present reference frame (time t) with a Spatial Transformer module S

Why Alignment Matters

Without warping to a common frame, the temporal model must learn ego-motion AND dynamic tracking simultaneously — a much harder problem.

VPQ: 29.9 → 24.6 (−5.3). Second largest ablation impact after removing temporal context entirely (−5.8).



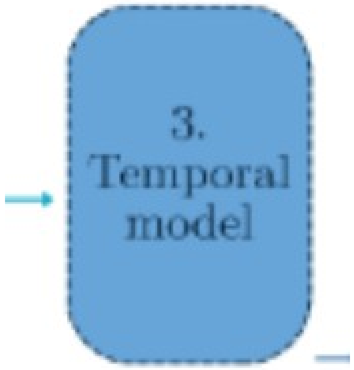
3. Temporal Model

3D convolutional temporal model learns a spatio-temporal state s_t.

Temporal Modelling Design

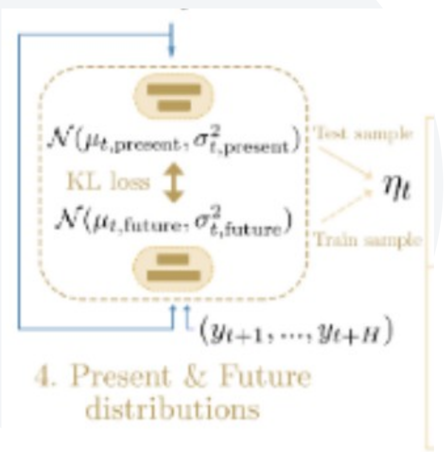
Prior work naively fed unaligned past frames (Luc et al., 2017) or concatenated features without temporal modelling (Fishing Net, 2020).

FIERY shows BOTH alignment AND a high-capacity temporal model (3D convolutions with global pooling) are required. Neither alone suffices for spatio-temporal correspondences.



4. Present & Future Distributions

We parametrise two probability distributions: the present and the future distribution. The present distribution is con-
ditioned on the current state st, and the future distribution is conditioned on both the current state st and future labels
(yt+1, ..., yt+H).



5. Future Prediction

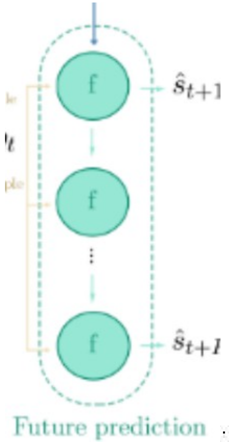
*We sample a latent code ηt from the future distribution during training, and from the present distribution during infer-
ence. The current state st and the latent code ηt are the inputs to the future prediction model that recursively predicts
future states (ˆst+1, ..., ˆst+H).*

Convolutional GRU Unrolling

Input: (s_t, η_t) → recursively predict future states: $\hat{s}_{t+1}, \hat{s}_{t+2}, \ldots, \hat{s}_{t+H}$

Structure: [Conv-GRU → 3× Residual Blocks (kernel 3×3)] repeated ×3

Sequential constraint: prediction at t+j+1 is conditioned on prediction at t+j. Ablation: VPQ −3.7 without unrolling.

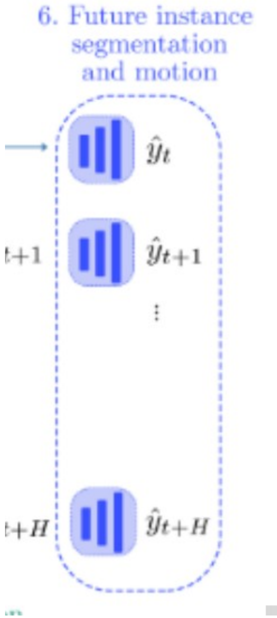


6. Future instance segmentation and motion

The states are decoded into future instance segmentation and future motion in bird's-eye view ($\hat{y}_t$, ..., $\hat{y}_{t+H}$).

Decoder: $D(\hat{s}_{t+j}) = \hat{y}_{t+j}$ for $j \in \{0, ..., H\}$, with $\hat{s}_t = s_t$

Segmentation $\in \mathbb{R}^{1\times H\times W}$	Binary vehicle occupancy map in BEV
Centerness $\in \mathbb{R}^{1\times H\times W}$	2D Gaussian heatmap at instance center of mass (σ=3) → NMS extracts centers
Offset $\in \mathbb{R}^{2\times H\times W}$	Vector field pointing each pixel to its instance center → assigns pixels to instances
Future flow $\in \mathbb{R}^{2\times H\times W}$	Displacement field between consecutive timesteps → Hungarian matching for tracking



Thank you for your attention!